Advanced Mathematics 1

Computing and Mathematics

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Advanced Mathematics 1 (A35244)

Short Title:	Advanced Mathematics 1		
Faculty	Science and Computing		
Department	Computing and Mathematics		
Cluster	Mathematics and Physics		
Author	ADAVY		
Credits	5	**Level:**	Introductory

Description of Module / Aims

The study of this module enables students to systematically master the basic concepts, basic theories and basic operation skills such as the limit of functions, the differential calculus of unary functions and its application, and the integral calculus of unary functions. Cultivate students' abstract thinking and abstract thinking ability, logical reasoning ability, and certain mathematical modeling ability, correctly understand some important mathematical thinking methods.

Programmes

`MATH-0030`	BEng (Hons) in Electrical and Automation Engineering (International) (`WD_ETRIC_BI`)	1 / 1 / M
`MATH-0030`	BEng (Hons) in Information Engineering (International) (`WD_EEELC_BI`)	1 / 1 / M
`MATH-0030`	BSc (Hons) in Applied Computing (International) (`WD_KACMI_B`)	1 / 1 / M
`MATH-0030`	BSc (Hons) in Software Engineering (`WD_KDEVP_BI`)	1 / 1 / M
`MATH-0030`	BSc (Hons) in the Internet of Things (International) (`WD_KINTT_BI`)	1 / 1 / M

Indicative Content

- Functions: polynomials; rational functions; transcendental functions; limits and continuity
- Differential calculus:the concept of derivative, the rule of derivative of unary function, higher order derivative, implicit function and derivative of parametric function; differentiation of unary functions and its applications; curve--sketching using critical points and simple problems of optimization; Newton's method; Taylor expansions
- Integral calculus: concepts, properties and integral methods of indefinite and definite integrals; the fundamental theorem of calculus; applications of definite integrals
- Vector algebra: vectors and their linear operations; vector scalar product, vector product and mixed product

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explain the most important properties of the elementary functions (including piecewise function) and illustrate these properties using appropriate examples.
2. Compute routine calculations with numbers and master a new mathematical idea and method to solve the problem- -limit method, understand the founding process of calculus.
3. Calculate the derivative and the integral of unary functions.
4. Use appropriate derivative and integral methods to solve elementary optimization problems.
5. Master the new idea of using algebraic methods to solve geometric problems.
6. Apply concepts indicated in the syllabus content to appropriate problems and interpret the solutions obtained.

Learning and Teaching Methods

- Delivery of the module will be through lectures, tutorials, discussion, and practice.
- The lectures will enable students to obtain the basic concepts, basic theories and basic operation skills of unary functions, and acquire the limit and continuity of unary functions, calculus of unary functions and its applications, vector algebra and other aspects, and will develop theory, lead students through worked examples and introduce the context for the module material.
- The tutorial classes will underpin and rehearse the skills covered in lectures. Students will be encouraged, through the problem sheets, to construct valid and precise mathematical arguments and will be expected to produce solutions using appropriate mathematical notation.
- The discussion and practical sessions will be used to discuss applications of the theory and to utilize and implement mathematical software.
- The practical programme is designed to re-enforce the strong interconnections between the mathematical concepts covered in this module and Computer Science concepts using Computer Algebra Systems (CAS) and guided development of students' own code. A typical activity consists of investigating a number of basic quadrature techniques and their application to problems in computer science.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	40%	1,2,3,4,5
Assignment	20%	1,2,3,4,5
In-Class Assessment	20%	1,2,3,4,5
Final Written Examination	60%	1,2,3,4,5,6

Assessment Criteria

<40%: Inability to understand the concept of functions, complete simple drill exercises in differentiation and integration.

40%-49%: Able to demonstrate a basic understanding of the fundamental concepts in calculus as outlined in the indicative content, and apply, to some degree, these concepts to problems in computer science.

50%-59%: All the above, in addition to using the standard notation to express mathematical entities.

60%-69%: In addition, to be able to determine appropriate mathematical techniques to analyze applied problems and to express their work with rigor and precision.

70%-100%: All the previous to an excellent level. Demonstrates an ability to put a solution into a context and assess whether such solutions are meaningful.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	96	

Supplementary Material

- "Helping Engineers Learn Mathematics (HELM)." Helping Engineers Learn Mathematics (HELM). https://www.lboro.ac.uk/departments/mlsc/student-resources/helm-workbooks/
- Banner, A. _Princeton Calculus Reader_. China: People's Posts and Telecommunications Press, 2016.

- Croft, A. and R. Davison. _Foundation Maths. 5._. New York: Pearson, 2010.
- Hughes-Hallett, D. _Calculus: Single and Multivariable._. 6th Edition. New York: Wiley, 2012.
- Shunfeng, W., Z. Jian, W. Yajuan and C. Lijuan. _Advanced Mathematics Guidance (__Volume 1)_. China: Southeast University Press, 2019.

Requested Resources

- COMPUTER LAB: Multimedia Lab